如何做应用型的研究?

倪明选
香港科技大学讲座教授
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大纲

- 什么是应用型研究?
- 做研究前需要准备那些功力?
- 如何去找一个好的研究题目？
- 如何把一个研究题目做好?
- 如何把一篇文章写好？
- 什么是理想的应用型研究?
- 结论

什么是研究(research)?

- 研究是 **Re (repeat) + search** 吗？
- 资料收集是研究吗？
- 事实的发现(**fact discovery**)是研究吗？
- 研究是创造新的知识
- 研究并没有一个简单的模式
  - 做研究的方法会因人而异或因事而异
  - 不同的人做研究的方法也可能不同
  - 用我自己超过30年的经验给大家参考

什么是理论型研究（Theoretical Research)？

- **信息领域与物理，数学等领域是不同的**
  - 理论物理，实验物理，计算物理
  - 纯数学，应用数学，计算数学
- **信息科学基本上是应用科学，也需理论的研究基础**
- **什么是信息科学的理论研究？**
  - NP-complete, graph theory, game theory, coding theory, information theory, etc.
  - 真正天才做的研究，例如清华的姚班
  - 研究结果不一定有直接应用价值

什么是应用型研究（Applied Research)？

- 研究的结果可能有实际的应用价值
- 什么是可能？
  - 短期或长期或不可预知的未来
- 时机到了吗？
- 需要的支撑技术成熟了吗？
- 解决的问题有意义吗？
- 解决问题的方法可行吗？

大部分文章初看又有理论又有应用，
但理论没有新意，应用没有价值！

科研的意义

99.99%的论文根本就没人看过

或者只有 4 个人看过（谁呢？）

99%的论文，就是好论文，也无法实用

科研就是浪费生命，浪费资源的无意义的行为？

科研工作，是探路；而科研工作者，是先驱；

全世界最聪明的人，用自己大量的时间，精力，物力，去帮助后人探索道路，告诉大家，此路通(或者不通)！

什么是好的研究

- **Results, positive or negative, are useful to others**
  - Original, reproducible, and available to others
  - Success is judged by the extent to which the results are quoted by other researchers.
- **Bad research is a dead-end that leads nowhere.**
- **Usually there is no way of assuring a successful outcome at the beginning of a research project.**
  - We may not be able to understand the significance of our results, or we may not get any results if an experiment proves more difficult than expected or a theory is found to be unsound.

对应用型研究的基本评估

- 提出的技术方案正确吗？
- 在直觉上是合理吗?
- 做出的成果有发表价值吗？在那里可发表?
- 做出的成果有专利的价值吗？可解决那些实际的应用？
- 做出的成果是否超出我们已知的知识?
- 做出的成果在那些领域会有影响力?
- …

做好的研究对你可能很难

- 因为你准备不够
- 因为你对研究的看法不对
- 因为你不知道如何去找问题，如何去描述一个问题
- 因为你没有经过创新的思考
- 因为你读的不够
- 因为你不知道利用网络或图书馆帮你寻找知识

科研所需要的基本素质

开放的思维

没事就爱找茬

穷追不舍的精神

认清形势，搞明白所有细节

越挫越勇的勇气

痛，并快乐着

忍受孤独

科研的路上，很可能只有你一个人

需要学习多久才能开始做研究

- **I have learnt all the mathematics, and I am loaded**
  - Discrete algorithms, partial differential equations, dynamic control, probabilistic modeling, information theory, etc.
- **I have read all papers out there from journals and conferences**
- **I still don't have a clue what to do in research.**
- **Where in the world is research topic?**
- **Can I do research now？**
  - There is no way you can cope with all of them
  - Majority of the published papers are junks, and can cause brain damage and can be misleading

你具备了做研究的基础么？

- **Logical thinking, af ter all we are in engineering world**
- **Basic skills**
  - E.g., in CSE, Algorithms, Complexity, Analytical skills, Programmi ng skills, etc.
- **Ability to learn**
  - Life long learning process, esp. in CSE
- **Abstraction - take a problem, you have to know**
  - What is/are the fundamental problem(s)?
  - You have to s ee both “forest and trees”
  - What have been done, why?
  - What are seemingly undoabl e?
- **Understand your strength and weakness**

你是否可以开始做研究了？

- **Open mind**
  - We are not dealing with math problems in that there exists perfect solutions
  - Engineering solutions are subject to argument and debate, i.e., each solution is a trade-off, and it only works in a constrained environment
- **Critical mind**
  - When you read others, it is equally important to understand what circumstance that it does not work as in which it works
  - What assumptions they made?
  - If you cannot identify such scenario, you are not understanding the problem
- **Finding a problem is more important, and difficult than solving a problem**
- **You need out-of-box thinking**

需要读多少论文？需要读什么论文？

- **Top conferences or workshops first**
  - ACM Sigcomm, ACM Mobicom, IEEE Infocom
  - IEEE ICNP, IWQoS, MobiHoc
- **Second tier conferences only for reference**
  - IEEE Globecom, ICC
- **Avoid bad conferences**
  - Regional, and less reputable ones
- **Look for some known researchers web sites**
- **Read journal papers only it has not been published else where, or when it contains more det ailed and complete treatment**

如何读一篇论文

- 这篇文章解决了什么问题？
- 为什么要解决这个问题？解决的问题有意义吗？
- 为什么别人没想到这个问题？
- 解决的方法够好吗？评估的方法合理吗？
- 别人有类似的方法吗？
- 文章的罩门在那里？
  - 用了那些假设？
  - 什么条件下不能适用？
- 有改良的空间吗？

研究题目该如何去找

- **Drink a beer, relax, and ideas will come to you**
- **The ideas fall from the sky!**
- **Understanding the existing work, build upon that incrementally**
  - A thorough understanding of those is the basis
- **Ideas in most cases come from the deep understanding of a subject, and possess of broad knowledge**
  - This is not a technical training, i.e., this is not about solving a bipartite graph, or differential equations
  - This is about relating them to real world problems
  - This is about providing new insight beyond known
  - This is about your creativity!

研究的不同类型

- **New problem and new solution**
  - This comes rarely, perhaps something you could only wish, once a life-time experience (such as Shannon theory)
- **Old problem and old solution**
  - Plenty of out there under the category of “Yet another paper on … ”
- **New problem and old solution/Old problem and new solution**
  - You may target the above two
  - Examples: new routing algorithms, a new branch of sub-problems

如何去找研究题目

- 从工程问题中去找出科学问题
  - Each solution to an engineering problem is only a trade-off; it is not a cure for all, it definitely has side-effect.
  - Example: P2P computing
- 集中**(focus)!** 研究面不要打的太宽，太分散
  - It should have a clear boundary
  - Focus on what can be done
- **Don't lose the bigger picture**
  - Tree and the forest
  - How does it help to solve one or more pieces in the bigger problem

研究题目的时间周期

- **It varies from days to years depending on the scope of research**
- **An early breakthrough in a piece of research may take a lengthy period to refine it**
- **It may take years of exploration before a breakthrough is made**
- 在工程性强的信息领域，研究的时间通常不会拖的很长

博士的典型科研

- **Make a plan earlier, for 3-4 years**
  - Identify a general category of problems
  - Your interest and your advisor's interest
- **Brainstorming with your team members**
- **Pick up a good problem at the beginning is important**
- **The research topics must be of current interest**
- **It has to be something that within your capability**
  - You need to understand your strength and weakness, and be realistic (don't shoot stars)
  - You should know your interest, self-motivation is one of the single most important factors

博士的研究生涯

- **Understand the basics, fundament al problems, and open issues**
- **Do I have to read all related papers?**
  - Read top 10 or 20 papers in the area first
  - Don’t get brainwashed easily
- **Think and read**
  - Put all papers into perspective
- **Start from a small yet concrete problem**
  - Build you skill and confidence
- **Discussion generates ideas**

选择一个研究方向

- **Before beginning any research**
  - Look around and do background seeking
  - Look in referenc e books or on the Internet for a topic that interest you and would present a good topic to research
- **Make sure you fully underst and what the question is and what you expect to find.**
- **Be direct, simple and clear when stating the research problem.**
- **Limit your study to a manageable size**
- **Limited time for PhD research**
  - But don't forget two important adjectives: "original" and "substantial"

如何把一个研究题目做好

- 把问题定义想清楚，讲清楚
- 把问题要解决的目标目的讲清楚
- 如果问题太大，把问题有机的分成几个小问题，一个一个来做
- 在研究中，可以给一些合理的假设（也是罩门所在）
- 你知道可能的解决方法吗？方法够实际吗？
- 多听听不同领域的高手演讲
- 有类似的解决方法吗？与其他的方法有比较吗？

如何把一个题目做好？

- 如何定义什么叫好？
- 如何去评估一个好的方法？
  - 做个原型机？用模拟？
  - 实验是检验真理最好的方法，虽然贵，值！
    - 例子： wireless sensor networks
- 在研究过程中，需要收集与分析各种数据
  - 数据可靠吗？
  - 数据合理吗？
- 答案如不够好，为什么？

典型论文的结构（科技八股）

- **Title**
  - A logical, accurate, descriptive and grammatically correct title
  - As short as possible
  - Two-part titles
- **Abstract**
  - Problem statement
  - Research approach
  - Summary of contributions
  - Easy to read and informative
- **Keywords**
  - Be specific
  - Use standard phrases (e.g., Computing Reviews)
- **Introduction**
  - What is the problem?
  - Why it is important?
  - Questions to be explored
  - Central idea(s) of the paper
  - Objectives
  - Paper organization
- **Related Work (or before Conclusion)**
  - Give credits to others
  - What have other people done?
  - Cite key papers

典型论文的结构（科技八股）

- **Your work (Main body)**
  - Algorithm/System/Theory
  - Detailed problem formulation
  - Evaluation metrics
  - Central concept underlying your work
  - Level of details
  - Illustrating example(s)
  - Complexity analysis
  - Answers to questions posed
  - Novel aspects of the work
  - Significance of the results
- **Performance Evaluation**
  - Describe results of experiments/simulations
  - Provide evidence in support of your work
  - Comparison with others
- **Conclusion and Future Work**
  - What was learned
  - How it can be applied
  - Possible future directions
- **References**
  - Standard format
- **Appendix**
  - Detailed protocol description, proofs, etc.

会议论文

- **Start earlier for a conference submission**
- **Which conference to submit?**
- **Deadline is the best drive for making progress**
- **What make a good paper: content and writing!**
- **Clear, convincing, simple and good English**
  - This is a never-ending optimization process, do this within the time and page limits
- **Review process 5/30 rule**
  - 5 minutes - Abstract, introduction, figure and conclusion
  - 30 minutes – understand 90% of the paper

期刊论文

- **A good conference paper (10%-25% acceptance rate) can be submitted to a journal, with 30% new results**
- **Report more compl ete and focused results**
- **Which journal to submit the paper?**
- **Give yourself a deadline**
- **Be patient with the long review and re-review**
- **At the earlier stage of one's career, don't quit if asked for major revision**
  - But don't do seemingly impossible

评审的过程

- **Low acceptance rate**
  - Typical in many CSE conferences
- **Reviewers are potential competitors**
  - Try to convince them and don’t give them excuses to kill the paper
- **Reviewers are very busy**
  - Try to make their job easier
- **English is not our strength**
- **Don’t try your luck, it won’t work!**

论文评审在看什么？

- **New problem or new solution?**
- **Are the main results significant?**
- **Is the paper technically correct?**
- **Does the paper provide a fair assessment of its strength and limitation?**
- **Is the paper clearly written, thus accessible to general readers?**
- **Are the references adequate?**
- **Is the paper appropriate for conference/journal?**
- **…**

常见的错误-A

- **Reviewers have to understand me**
- **Only you know your work well, not reviewers**
- **Make it easier to understand**
  - Motivation and rationales
  - Control the level of details
  - Make connection throughout the paper
  - Use examples, graph, flow chat whenever needed
  - Pose questions, and answer them

常见的错误-B

- **Formality leads to elegance**
- **Keep it simple, perhaps stupid**
- **Start with motivations and rationales**
- **Avoid unnecessary formality**
- **Avoid casual writing**
  - "believe me, this is really a good work"
- **Flow and logic is much more i mportant, proofreading does not help you with that**

常见的错误-C

- **<u>I have 10 contributions</u>**
  - Reviewers should see this in a masterpiece
- **Focus in the key**
  - One problem, one solution in a conference paper
  - One problem, more complete solutions in a journal paper
  - Thorough and deep
- **Emphasize but not exaggerate your contributions**
  - Say it is “significant” only if it is, and justify it
  - If it is the first time, say “This is, to the best of our knowledge, the first time … “

常见的错误-D

- **Reviewers are evil**
  - They reject paper so their papers can be accepted
  - They reject my paper, so to steal my ideas
- **Reviewers are critical**
- **You have to be a good salesman to convince them**
- **Don't give reviewers a chance to kill you (confess shortcomings)**

案例：Safe charging

Safe charging for **wireless power transfer**

H Dai, Y Liu, G Chen, X Wu, T He... - ... /ACM Transactions on ..., 2017 - ieeexplore.ieee.org

... to be **transferred** from a **charger** to ambient devices wirelessly... Existing schemes mainly focus on the **power transfer** ... study the **safe charging** problem (SCP) of scheduling **power** chargers ...

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- **Haipeng Dai:** Nearly no pub. in 3 years, and 10+ papers in later years
- **Not easily get brainwashed**
- **Critical thingk**
- **Branch out a new problem**

案例2：Route Prediction

Route prediction for **instant delivery**

Y Zhang, Y Liu, G Li, Y Ding, N Chen, H Zhang… - Proceedings of the …, 2019 - dl.acm.org

… , we studied the **route prediction** in the scene of **instant delivery**. The core part of **route prediction** is to … The χ2 test is used select the appropriate features and we design a **route prediction** …

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- **A MS. Student in three months**
- **Deep understanding of the problem**

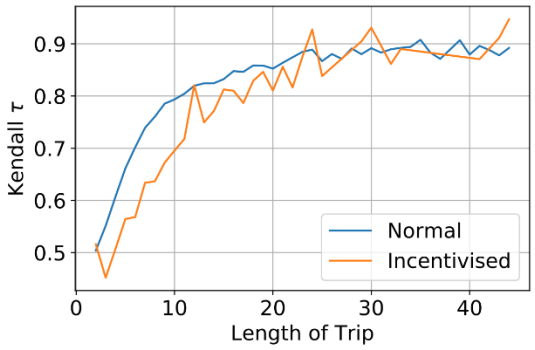

图3 推荐路径与骑手自选路径的差异图

案例3：LANDMARC

Opportunity-based topology control in wireless sensor networks

Y Liu, Q Zhang, L Ni

IEEE Transactions on Parallel and distributed systems, 2009 • ieeexplore.ieee.org

Topology control is an effective method to improve the energy efficiency of wireless sensor networks (WSNs). Traditional approaches are based on the assumption that a pair of nodes is either "connected" or "disconnected." These approaches are called connectivity-based topology control. In real environments, however, there are many intermittently connected wireless links called lossy links. Taking a succeeded lossy link as an advantage, we are able to construct more energy-efficient topologies. Toward this end, we

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- **Topology Control is a well studied problem**
- **Why is it still a problem?**

案例4：Transceiver-free tracking

Rass: A real-time, accurate, and scalable system for tracking transceiver-free objects

D Zhang, Y Liu, X Guo, LM Ni

IEEE Transactions on Parallel and Distributed Systems, 2012 • ieeexplore.ieee.org

Transceiver-free object tracking is to trace a moving object that does not carry any communication device in an environment with some monitoring nodes predeployed. Among all the tracking technologies, RF-based technology is an emerging research field facing many challenges. Although we proposed the original idea, until now there is no method achieving scalability without sacrificing latency and accuracy. In this paper, we put forward a real-time tracking system RASS, which can achieve this goal and is promising in

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- **Tracking without devices**

如何读好一篇文章

- 干什么的 (应用背景)
- 目标是什么(应用需求)
- 为什么目标达不到
- 前人怎么做的，为啥没做好
- 我们怎么做的，为啥能做的好
- 我们遇到什么困难了，怎么解决的
- 我们怎么证明能做的好(分析，实验)
- 做好笔记

要读多少文章

- 要读好文章，顶级会议
- 灌水文章，**Permanent Head Damage (PHD)**
- **CCF A** 类
- **Sigcomm, Mobicom, OSDI, NSDI**
- **10-20**篇，找到圈子里面的大牛，跟着他们的思路看文章
- 最近**1-2**年的
- 期刊仅供参考

如何寻找好的IDEA

那一踹的风情

“官吏用迅雷不及掩耳盗铃之势对准斛猛踹一脚！此时超出斛壁的部分谷粒会倒在地上，老百姓慌忙去捡，此时官吏会大声叫喊：别捡，那是损耗！喂，说你呢，还捡！

这就是淋尖踢斛

--当年明月 明朝

那些事”

如何寻找IDEA

IDEA 是从天上掉下来的

寻找题目时，要博览众家之长，多看，多想，全面撒网，重点捕捞

一旦题目定下来，就得Focus

- 讲究深度
- 一篇文章，通常只需要讲明白一件事情
- 就说一句话，让人家信服也不容易

什么是一个好IDEA

解决的问题要有意义

不是空穴来风，让人家觉得你是无病呻吟

问题要够难

不是人家一下子就能解决的，就对你的解决方案有兴趣看啦

问题不能太难

太难你做不出来

你能够验证你的方法确实解决了问题

实验，分析，仿真，实现

如何做好题目

问题定义想清楚，讲清楚

输入是什么，输出是什么

把问题要解决的目标,目的讲清楚

可量化指标，给定两个方法，一定能说出谁好谁不好

如果问题太大，把问题有机的分成几个小问题，一个一个来做

在研究中，可以给一些合理的假设（也是罩门所在）

假设太多，没意义

假设太少，做不出

多听听不同领域的高手演讲

实验是检验真理最好的方法，虽然贵，值！

实验➔理论 ➔实验

科研要作多久

- 可能很短，可能很长
- 你用了多少时间去证明你的IDEA,别人就需要多少时间去反驳你
- 灵感的闪现可能需要经年累月的工作证明
- 大胆假设，小心求证！
- 按照半年时间管理一个题目，瞄准一个会议先

学术道德问题

- **Plagiarism**
- **Falsifying data**
- **Double submission**
- **Self-plagiarism**
- **Do not steal others idea**
- **Give credits to others**
- **Do not take chance**

理想的应用研究

- 倪明选的四个能
  - 能对国家社会有贡献
  - 能够产生一流的文章
  - 能够培养一流的学生
  - 能够带来经济的利益
- 孙家广院士的四个得
  - 想得出
  - 做得到
  - 用得着
  - 卖得掉

结束语

- 做研究需要创新，有耐心，肯努力，肯坚持
- 培养好的说与写的能力
  - IQ cannot be improved, English can be improved
- 养成挑战（在知识方面）的性格
  - 在工程中，提出为什么？
- 跳出框框来想，不怕提出问题
- 不要闭门造车，与比你聪明的人及能与你互补的人一起合作
- 尽信书不如无书：不要轻易相信你读的
- 让做研究变成一种享受

致谢

- **This is my personal view**
  - Different researchers may have different views
- **I wish to thank all my students (past and present) and research collaborators for being part of my research career**
- **Some materials of this talk were based on talks by Charles X. Ling (Univ. of Western Ontario), Jim Kurose (UMass at Amherst), Bo Li (HKUST), Qiang Yang (HKUST), and thanks to Google**

Thank You！ 谢谢！

Q&A